Chapter 5—Eigenvalues, Eigenvectors and Invariant Subspaces

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Plan: Cover Sections A, C, D, E

§ 5.A Invariant Subspaces

Story/Me:

- The text uses the notion of direct sums
  that we have not seen so far.
- So, let us quickly look at its definition.

1.40 Definition ***direct sum***

Suppose $U_1, \ldots, U_m$ are subspaces of V.

- The sum $U_1 + \cdots + U_m$ is called a ***direct sum*** if each element of $U_1 + \cdots + U_m$ can be written in only one way as a sum $u_1 + \cdots + u_m$, where each u_j is in U_j.
- If $U_1 + \cdots + U_m$ is a direct sum, then $U_1 \oplus \cdots \oplus U_m$ denotes $U_1 + \cdots + U_m$, with the $\oplus$ notation serving as an indication that this is a direct sum.

- Here are a few examples

1.41 Example Suppose U is the subspace of $\mathbf{F}^3$ of those vectors whose last coordinate equals 0, and W is the subspace of $\mathbf{F}^3$ of those vectors whose first two coordinates equal 0:

$$U = \{(x, y, 0) \in \mathbf{F}^3 : x, y \in \mathbf{F}\} \quad \text{and} \quad W = \{(0, 0, z) \in \mathbf{F}^3 : z \in \mathbf{F}\}.$$

Then $\mathbf{F}^3 = U \oplus W$, as you should verify.

1.42 Example Suppose U_j is the subspace of $\mathbf{F}^n$ of those vectors whose coordinates are all 0, except possibly in the jth slot (thus, for example, $U_2 = \{(0, x, 0, \ldots, 0) \in \mathbf{F}^n : x \in \mathbf{F}\}$). Then

$$\mathbf{F}^n = U_1 \oplus \cdots \oplus U_n,$$

as you should verify.

1.43 Example Let

$$U_1 = \{(x, y, 0) \in \mathbf{F}^3 : x, y \in \mathbf{F}\},$$
$$U_2 = \{(0, 0, z) \in \mathbf{F}^3 : z \in \mathbf{F}\},$$
$$U_3 = \{(0, y, y) \in \mathbf{F}^3 : y \in \mathbf{F}\}.$$

Show that $U_1 + U_2 + U_3$ is not a direct sum.

- Here are some handy properties

1.44 Condition for a direct sum

Suppose $U_1, \ldots, U_m$ are subspaces of V. Then $U_1 + \cdots + U_m$ is a direct sum if and only if the only way to write 0 as a sum $u_1 + \cdots + u_m$, where each u_j is in U_j, is by taking each u_j equal to 0.

1.45 Direct sum of two subspaces

Suppose U and W are subspaces of V. Then $U + W$ is a direct sum if and only if $U \cap W = \{0\}$.

2.34 Every subspace of V is part of a direct sum equal to V

Suppose V is finite-dimensional and U is a subspace of V. Then there is a subspace W of V such that $V = U \oplus W$.

3.78 A sum is a direct sum if and only if dimensions add up

Suppose V is finite-dimensional and $U_1, \ldots, U_m$ are subspaces of V. Then $U_1 + \cdots + U_m$ is a direct sum if and only if

$$\dim(U_1 + \cdots + U_m) = \dim U_1 + \cdots + \dim U_m.$$

- Ok, so we can start the section

Story:

- The goal of this chapter is to develop tools
  that help us understand the structure of operators.

- Recall
  - An operator is a linear map
    from a vector space to itself.

  - We denote the set of operators on V by $\mathcal{L}(V) = \mathcal{L}(V, V)$.

- Suppose $T \in \mathcal{L}(V)$.
- If we have a direct sum decomposition
  $V = U_1 \oplus \cdots \oplus U_m$
  where each U_j is a proper subspace of V
  then to understand the behaviour of T
  we need only understand the behaviour of each $T|_{U_j}$

Notation.

$T|_{U_j}$denotes the restriction of T to the smaller domain U_j.

(returning to story)

- Dealing with $T_{|U_j}$ should be easie
  because it is a smaller vector space than V.

- However, if we intend to apply tools
  that we developed in the study of operators
  (such as taking powers)
  then we have a problem—$T_{|U_j}$ may not map U_j to itself

  (i.e. T restricted to U_j may not be an operator).

- This suggests that
  we should consider only decompositions of V
  of the form above
  where T maps each U_j to itself.

- This notion turns out to be quite important and therefore
  such spaces have their own name.

Definition 5.2

(Invariant subspace)

Suppose $T \in L(V)$.
A *subspace* U of V is called **invariant** under T
if
$u \in U$ implies $Tu \in U$.

Example 5.3

Suppose $T \in L(V)$.
Then each of the following subspaces of V are invariant under T
(a) $\{0\}$
(b) V
(c) $\text{null}(T)$
(d) $\text{range}(T)$

Story:

- Must an operator $T \in L(V)$
  have any invariant subspaces other than $\{0\}$ and V?

  Later we will see that the answer is yes—if V if finite dimensional
  and
  $\dim V > 1$ (for $F = C$) or
  $\dim V > 2$ (for $F = R$).

- Why don't the nullspace (T) or the range (T) suffice to answer this question?
  - Well, because null T may itself be equal to $\{0\}$ and
  - the range may be equal to V
    (this happens, for instance, when T is invertible)

Notation.

- We use $\mathcal{P}(R)$ to denote the vector space of polynomials over the field R
- We use $\mathcal{P}_4(R)$ to denote $\mathcal{P}(R)$ when restricted to polynomials of degree at most 4
- If p is a polynomial, p' is used to denote its derivative

Example 5.4

Suppose that $T \in L(\mathcal{P}(R))$ is defined by $Tp = p'$.
Then
$\mathcal{P}_4(R)$ (a subspace of $\mathcal{P}(R)$)
is invariant under T
because if $p \in \mathcal{P}(R)$ has degree at most 4
then
p' also has degree at most 4.

Eigenvalues and Eigenvectors

Story

- We return later to a deeper study of invariant subspaces.
- For now,
  we focus on the simplest possible non-trivial invariant subspaces
  subspaces with dimension 1.

- Take any $v \in V$ with $v \neq 0$ and
  let U equal the set of all scalar multiples of v:
  $U = \{\lambda v : \lambda \in F\} = \text{span}(v)$.

- Then U is a 1-dimensional subspace of V <--- maybe HW?
  (and every 1-dimensional subspace of V is of this form
  for an appropriate choice of v).

- If U is invariant under an operator $T \in L(V)$
  then $Tv \in U$,
  and
  hence there is a scalar $\lambda \in F$ such that
  $Tv = \lambda v$.

- Conversely, if $Tv = \lambda v$ for some $\lambda \in F$
  then
  $\text{span}(v)$ is a 1-dimensional subspace of V
  invariant under T.

- The equation $Tv = \lambda v$
  is quite important and has specific names associated with it

Definition 5.5
(Eigenvalue)

Suppose $T \in L(V)$.
A *number* $\lambda \in F$ is called an **eigenvalue** if T
if there exists $v \in V$ such that $v \neq 0$ and $Tv = \lambda v$.

Remark:
It is important to require that $v \neq 0$
otherwise every scalar $\lambda \in F$
satisfies $T0 = \lambda 0$.

NB: Based on the discussion obove we have that
T has a 1-dimensional invarinate subspace
iff
T has an eigenvalue.

Proposition 5.6 (Equivalent conditions to be an eigenvalue)

Suppose
V is finite-dimensional,
$T \in L(V)$ and
$\lambda \in F$.

Then
the following are equivalent:

(a) λ is an eigenvalue of T
(b) $T - \lambda I$ is not injective
(c) $T - \lambda I$ is not surjective
(d) $T - \lambda I$ is not invertible.

Proof.
Conditions (a) and (b) are equivalent because
$Tv = \lambda v$ is equivalent to $(T - \lambda I)v = 0$.

Conditions (b), (c) and (d) are equivalent
(using Theorem 9, Chapter 3, Hoffman & Kunze)
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Theorem 9.
Let V and W be finite dimensional vector spaces over the field F
such that $\dim V = \dim W$.

If T is a linear transformation from V into W
the following are equivalent.

(i) T is invertible
(ii) T is non-singular
(iii) T is onto (i.e. range of T is W).

Definition 5.7 (eigenvector)

Suppose $T \in L(V)$ and $\lambda \in F$ is an eigenvalue of T.
A *vector* $v \in V$ is called an
eigenvector of T
corresponding to λ if
$v \neq 0$ and $Tv = \lambda v$.

Remark.

- Since $Tv = \lambda v$ iff $(T - \lambda I)v = 0$, it holds that

  a vector $v \in V$ with $v \neq 0$ is an eigenvector of T
  corresponding to λ
  iff
  $v \in \text{null}(T - \lambda I)$.

Illustration.

Suppose $T \in L(F^2)$ is defined by
$T(w, z) = (-z, w)$.

(a) Find the eigenvalues and eigenvectors of T if $F = R$.
(b) Find the eigenvalues and eigenvectors of T if $F = C$.

(a) [this is an intuitive argument]

When $F = R$
T can be seen as a counterclockwise rotation
by 90° about the origin in R^2.

An operator has an eigenvalue
iff
there is a nonzero vector in its domain
that gets sent by the operator
to a scalar multiple of itself.

A 90° rotation of a nonzero vector in R^2
is "obviously" never equal to a scalar multiple of itself.

Thus, if $F = R$
T has no eigenvalues (and thus no eigenvectors).

[the formal argument can be adapted from how we show part b]

(b) To find eigenvalues of T

we must find scalars λ such that
$T(w, z) = \lambda(w, z)$
has some solution
other than $w = z = 0$.

The equation above is equivalent to the simultaneous equations
$-z = \lambda w$, and $w = \lambda z$. [5.9]

Substituting the value for w
given by the second equation into the first equation
gives
$-z = \lambda^2 z$.

Now, z cannot equal 0
(otherwise 5.9 would say $w = 0$
and we want a non-zero vector)

Therefore, the equation above leads to the equation
$-1 = \lambda^2$.

The solutions to this equation are $\lambda = i$ and $\lambda = -i$.

One can easily verify that i and $-i$ are eigenvalues of T.
To see this
simply construct the eigenvectors using [5.9]
to obtain
$(w, -wi)$ as the eigenvector for eigenvalue i

(w, wi) as the eigenvector for eigenvalue $-i$.
(where $w \in C$ is an arbitrary complex number).

Story

- We now show that eigenvectors corresponding to
  distinct eigenvalues are
  linearly independent.

Proposition 5.10 (Linearly independent eigenvectors)

Let $T \in L(V)$.
Suppose
$\lambda_1, \dots \lambda_m$ are distinct eigenvalues of T and
$v_1, \dots v_m$ are the corresponding eigenvectors.

Then,
$v_1, \dots v_m$ are linearly independent.

Proof.

We prove by contradiction.

Suppose $v_1, \dots v_m$ are linearly dependent.

Let k be the smallest positive integer such that
$v_k \in \operatorname{span}(v_1, \dots v_{k-1})$ [5.11]

Thus,
there exist $a_1, \dots a_{k-1} \in F$ such that
$v_k = a_1 v_1 + \cdots + a_{k-1} v_{k-1}$ [5.12]

Apply T to both sides of this equation
to get
$\lambda_k v_k = a_1 \lambda_1 v_1 + \cdots + a_{k-1} \lambda_{k-1} v_{k-1}$.

Multiply both sides of 5.12 by λ_k and
subtract from the equation above
to get

$$0 = a_1(\lambda_k - \lambda_1) v_1 + \cdots + a_{k-1}(\lambda_k - \lambda_{k-1}) v_{k-1}.$$

Since we chose k to be the smallest positive integer
satisfying [5.11]
it follows that
$v_1, \dots v_{k-1}$ are linearly independent.

Thus the equation above means that
all a's must be zero
(recall that λ_k is not equal to
any of the $\lambda_1, \dots \lambda_{k-1}$ by the premise)

This in turn, using [5.12], entails that $v_k = \vec{0}$.

But this contradicts our hypothesis that
v_k is an eigenvector.

Thus, our assumption that $v_1, \dots v_m$ is linearly dependent
must be false.

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Story:

- The corollary below states that
  an operator cannot have more distinct eigenvalues
  than
  the dimension of the vector space on which it acts.

Corollary 5.13 (Number of eigenvalues)
Suppose V is finite-dimensional.
Then each operator on V
has at most $\dim V$ distinct eigenvalues.

Proof.
Suppose $T \in L(V)$ has distinct eigenvalues $\lambda_1, \dots \lambda_m$.
Let $v_1, \dots v_m$ be the corresponding eigenvectors.
It follows from Proposition 5.10 that
$v_1, \dots v_m$ are linearly independent.
Thus $m \leq \dim \mathrm{V}$ (this in turn follows from our discussions from Hoffman and Kunze).

■

Story.
- If $T \in L(V)$
  and
  U is a subspace of V
  invariant under T
  then
  U determines two other operators
  $T_{|U} \in L(U)$ and $T/U \in L(V/U)$ in a natural way

Definition 5.14 ($T_{|U}$ and T/U)
Suppose $T \in L(V)$ and U is a subspace of V invariant under T.
- The **restriction operator** $T_{|U} \in L(U)$ is defined by
  $T_{|U}(u) = Tu$
  for all $u \in U$.
- The **quotient operator** $T/U \in L(V/U)$ is defined by
  $(T/U)(v + U) = Tv + U$
  for all $v \in V$.

Story.
- For both operators defined above
  let us take a moment to ensure they are well defined as operators on their domains.
- First
  consider the restriction operator $T_{|U}$
  which is T with its domain restirted to U
  thought of as a mapping into U instead of V.

  The condition that U is invariant under T
  is what allows us to think of $T_{|U}$ as an operator on U
  (i.e. a linear map into the same space as its domain
  rather than simply a linear map
  from one vector space to another)

- To see that the definition for the quotient operator makes sense
  one needs to verify that
  if $v + U = w + U$
  then
  $Tv + U = Tw + U$.

  To this end, suppose that $v + U = w + U$.
  This means that $v - w \in U$.
  (from Statement 3.85).

  Now, since U is invariant under T
  it holds that $T(v - w) \in U$.
  That, in turn means that

Proof is HW

3.85 Two affine subsets parallel to U are equal or disjoint

Suppose U is a subspace of V and $v, w \in V$. Then the following are equivalent:

(a) $v - w \in U$;

(b) $v + U = w + U$;

$Tv - Tw \in U$
which is equivalent to saying
$Tv + U = Tw + U$ as desired.

(c) $(v + U) \cap (w + U) \neq \varnothing$.

- How is all this used?
  - Suppose T is an operator on finite-dimensional vector space V
    and U is a subspace of V invariant under T
    with $U \neq \{\vec{0}\}$ and $U \neq V$.

    In some sense,
    one can learn about T by studying the operators
    $T_{|U}$ and T/U
    each of which is an operator on
    a vector space with smaller dimension than V.

    (we will see this in statement 5.27)

  - On the other hand, sometimes $T_{|U}$ and T/U do not provide
    enough informatino about T.
    In the next example
    both $T_{|U}$ and T/U are 0
    even though T is not the 0 operator.

Example 5.15

Define as an operator $T \in L(F^2)$ by $T(x, y) = (y, 0)$.
Let $U = \{(x, 0): x \in F\}$.

Show that

(a) U is invariant under T and $T_{|U}$ is the 0 operator on U;
(b) there does not exist a subspace W on F^2 that is
invariant under T and is such that $F^2 = U \oplus W$;
(c) T/U is the 0 operator on F^2/U.

Solution.

(a) For $(x, 0) \in U$
we have $T(x, 0) = (0,0) \in U$.
Thus U is invariant under T and
$T_{|U}$ is the 0 operator on U

(b) Suppose W is a subspace of V
such that $F^2 = U \oplus W$.

Since $\dim F^2 = 2$ and $\dim U = 1$
we have that $\dim W = 1$.

If W were invariant under T
then each non-zero vector in W
would be an eigenvector of T.

Claim. All eigenvectors of T are in U. <--- can be considered HW
Why?
Note that 0 is the only eigenvalue of T
and
all eigenvectors of T are in U.

Thus, W is not invariant under T.

(c) For $(x, y) \in \mathbf{F}^2$, we have

$$\begin{aligned}(T/U)\big((x, y) + U\big) &= T(x, y) + U \\ &= (y, 0) + U \\ &= 0 + U,\end{aligned}$$